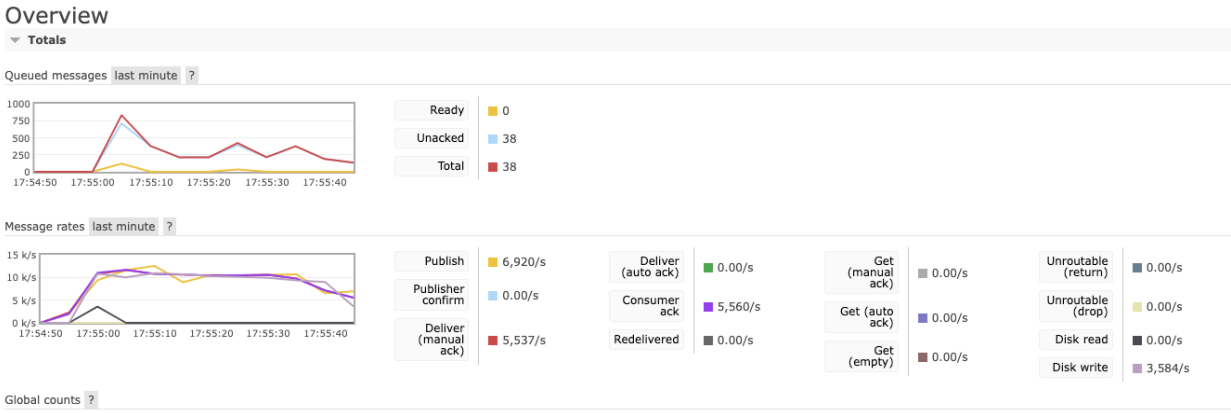
Test 1: 500K Baseline:

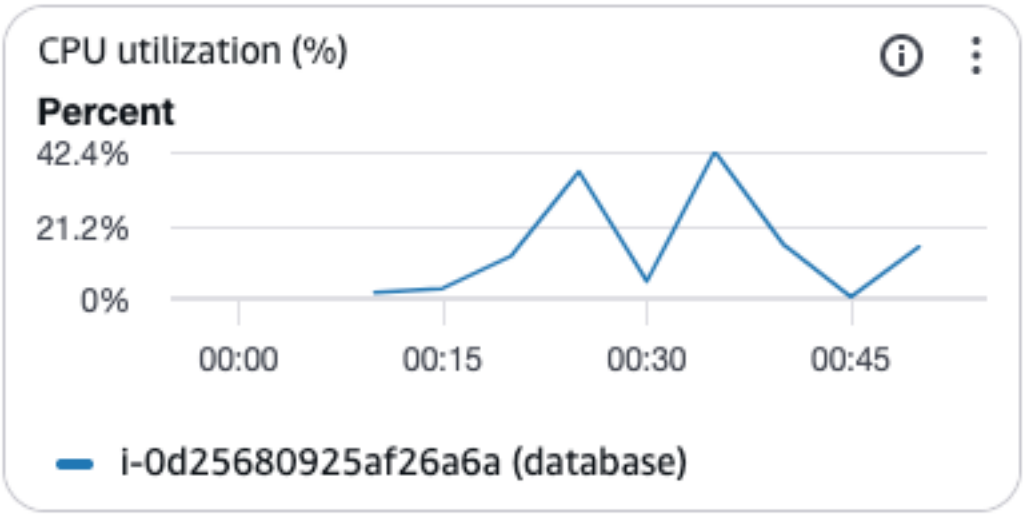
```
[ec2-user@ip-172-31-29-109 ~]$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "totalMessages": 500000,
    "activeUsersLastHour": 99325,
    "messagesByType": [
        {
            "_id": "TEXT",
            "count": 166149
        },
        {
            "_id": "JOIN",
            "count": 167335
        },
        {
            "_id": "LEAVE",
            "count": 166516
        }
    ],
    "topUsers": [
        {
            "_id": "61609",
            "messageCount": 17
        },
        {
            "_id": "32466",
            "messageCount": 16
        },
        {
            "_id": "47746",
            "messageCount": 16
        },
        {
            "_id": "67080",
            "messageCount": 16
        },
        {
            "_id": "97446",
            "messageCount": 16
        },
        {
            "_id": "1016",
            "messageCount": 15
        },
        {
```

```
        },
        {
            "_id": "14051",
            "messageCount": 15
        }
    ],
    "topRooms": [
        {
            "_id": "15",
            "messageCount": 26224
        },
        {
            "_id": "6",
            "messageCount": 26152
        },
        {
            "_id": "16",
            "messageCount": 25686
        },
        {
            "_id": "7",
            "messageCount": 25626
        },
        {
            "_id": "19",
            "messageCount": 25613
        },
        {
            "_id": "1",
            "messageCount": 25420
        },
        {
            "_id": "12",
            "messageCount": 25239
        },
        {
            "_id": "10",
            "messageCount": 25227
        },
        {
            "_id": "14",
            "messageCount": 25201
        },
        {
            "_id": "5",
            "messageCount": 25168
        }
    ],
    "room1MessageCount": 25420,
    "dbWriter": {
        "written": 0,
        "duplicates": 0,
        "failed": 0,
        "batches": 0,
        "bufferSize": 0
    },
    "totalQueryTimeMs": 3876
}
```

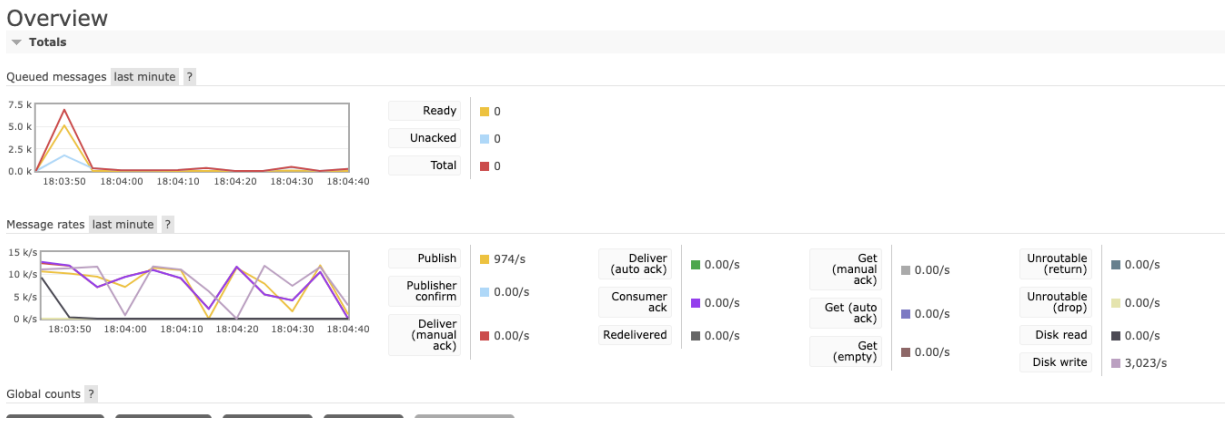
```
All messages sent. Waiting for responses...

==========================================
         Statistical Analysis Report
==========================================
Total Duration:      122996 ms
Messages Sent:       500000
Messages Received:   500000
Messages Failed:     0
Delivery Rate:       100.0%
Total Throughput:    4065.17 msg/sec
------------------------------------------
Mean Latency:        24999.67 ms
Median Latency:      24012 ms
95th % Latency:      51434 ms
99th % Latency:      57897 ms
Min Latency:         100 ms
Max Latency:         58154 ms
------------------------------------------
Message Distribution:
  TEXT:  166149
  JOIN:  167335
  LEAVE: 166516
------------------------------------------
Throughput Per Room:
  Room  1: 206.67 msg/sec
  Room  2: 197.06 msg/sec
  Room  3: 196.26 msg/sec
  Room  4: 196.74 msg/sec
```


Overview
Totals
Queued messages last minute ?
1000
750
500
250
0
17:54:50 17:55:00 17:55:10 17:55:20 17:55:30 17:55:40
Ready 0
Unacked 38
Total 38
Message rates last minute ?
15 k/s
10 k/s
5 k/s
0 k/s
17:54:50 17:55:00 17:55:10 17:55:20 17:55:30 17:55:40
Publish 6,920/s
Publisher confirm 0.00/s
Deliver (manual ack) 5,537/s
Deliver (auto ack) 0.00/s
Consumer ack 5,560/s
Redelivered 0.00/s
Get (manual ack) 0.00/s
Get (auto ack) 0.00/s
Get (empty) 0.00/s
Unroutable (return) 0.00/s
Unroutable (drop) 0.00/s
Disk read 0.00/s
Disk write 3,584/s
Global counts ?


CPU utilization (%)
Percent
42.4%
21.2%
0%
00:00
00:15
00:30
00:45
i-0d25680925af26a6a (database)

Test 2: 1M Stress Test:

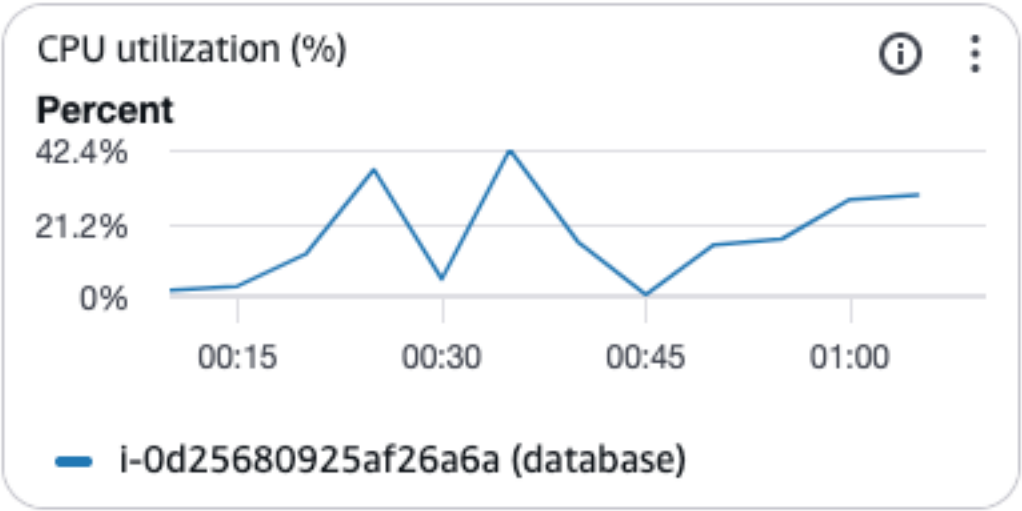


```
WARNING: No new responses for 120s, stopping wait.

==========================================
         Statistical Analysis Report
==========================================
Total Duration:      245554 ms
Messages Sent:       1000000
Messages Received:   966234
Messages Failed:     0
Delivery Rate:       96.6%
Total Throughput:    3934.91 msg/sec
------------------------------------------
Mean Latency:        55782.68 ms
Median Latency:      54259 ms
95th % Latency:      114171 ms
99th % Latency:      120088 ms
Min Latency:         55 ms
Max Latency:         120818 ms
------------------------------------------
Message Distribution:
  TEXT:  322136
  JOIN:  321965
  LEAVE: 322133
------------------------------------------
Throughput Per Room:
  Room  1: 171.09 msg/sec
  Room  2: 197.41 msg/sec
  Room  3: 200.49 msg/sec
  Room  4: 204.64 msg/sec
  Room  5: 204.20 msg/sec
  Room  6: 208.56 msg/sec
  Room  7: 203.73 msg/sec
  Room  8: 194.54 msg/sec
```

```
[ec2-user@ip-172-31-29-109 ~]$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "totalMessages": 904232,
    "activeUsersLastHour": 99990,
    "messagesByType": [
        {
            "_id": "JOIN",
            "count": 301421
        },
        {
            "_id": "LEAVE",
            "count": 301410
        },
        {
            "_id": "TEXT",
            "count": 301401
        }
    ],
    "topUsers": [
        {
            "_id": "57944",
            "messageCount": 24
        },
        {
            "_id": "2601",
            "messageCount": 24
        },
        {
            "_id": "68739",
            "messageCount": 23
        },
        {
            "_id": "41699",
            "messageCount": 23
        },
        {
            "_id": "60193",
            "messageCount": 23
        },
```

```
    ],
    "topRooms": [
        {
            "_id": "17",
            "messageCount": 47907
        },
        {
            "_id": "6",
            "messageCount": 47683
        },
        {
            "_id": "4",
            "messageCount": 47346
        },
        {
            "_id": "9",
            "messageCount": 47227
        },
        {
            "_id": "5",
            "messageCount": 46968
        },
        {
            "_id": "18",
            "messageCount": 46846
        },
        {
            "_id": "7",
            "messageCount": 46728
        },
        {
            "_id": "11",
            "messageCount": 46192
        },
        {
            "_id": "20",
            "messageCount": 46123
        },
        {
            "_id": "19",
            "messageCount": 45857
        }
    ],
    "room1MessageCount": 39558,
    "dbWriter": {
        "failed": 0,
        "duplicates": 0,
        "written": 904232,
        "bufferSize": 0,
        "batches": 33152
    },
    "totalQueryTimeMs": 9241
}
```


CPU utilization (%)
Percent
42.4%
21.2%
0%
00:15
00:30
00:45
01:00
i-0d25680925af26a6a (database)

Batch Test

100/500ms

```
==========================================
         Statistical Analysis Report
==========================================
Total Duration:      122457 ms
Messages Sent:       100000
Messages Received:   100000
Messages Failed:     0
Delivery Rate:       100.0%
Total Throughput:    816.61 msg/sec
------------------------------------------
Mean Latency:        4017.33 ms
Median Latency:      3883 ms
95th % Latency:      7571 ms
99th % Latency:      7891 ms
Min Latency:         63 ms
Max Latency:         7953 ms
------------------------------------------
Message Distribution:
  TEXT:  33293
  JOIN:  33318
  LEAVE: 33389
------------------------------------------
Throughput Per Room:
  Room  1: 41.35 msg/sec
  Room  2: 39.56 msg/sec
  Room  3: 36.89 msg/sec
  Room  4: 40.03 msg/sec
  Room  5: 41.23 msg/sec
  Room  6: 39.89 msg/sec
  Room  7: 42.13 msg/sec
```

```
[ec2-user@ip-172-31-29-109 ~]$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "totalMessages": 100000,
    "activeUsersLastHour": 63104,
    "messagesByType": [
        {
            "_id": "TEXT",
            "count": 33293
        },
        {
            "_id": "JOIN",
            "count": 33318
        },
        {
            "_id": "LEAVE",
            "count": 33389
        }
    ],
    "topUsers": [
        {
            "_id": "37623",
            "messageCount": 9
        },
        {
            "_id": "30289",
            "messageCount": 8
        },
        {
            "_id": "47628",
            "messageCount": 8
        },
        {
            "_id": "84428",
            "messageCount": 7
        },
        {
            "_id": "9378",
            "messageCount": 7
        },
        {
            "_id": "29788",
            "messageCount": 7
        },
        {
            "_id": "65838",
            "messageCount": 7
        },
        {
```

```
    },
    "topRooms": [
        {
            "_id": "10",
            "messageCount": 5268
        },
        {
            "_id": "20",
            "messageCount": 5215
        },
        {
            "_id": "12",
            "messageCount": 5179
        },
        {
            "_id": "17",
            "messageCount": 5176
        },
        {
            "_id": "7",
            "messageCount": 5159
        },
        {
            "_id": "16",
            "messageCount": 5103
        },
        {
            "_id": "14",
            "messageCount": 5098
        },
        {
            "_id": "8",
            "messageCount": 5090
        },
        {
            "_id": "19",
            "messageCount": 5080
        },
        {
            "_id": "1",
            "messageCount": 5063
        }
    ],
    "room1MessageCount": 5063,
    "dbWriter": {
        "failed": 0,
        "duplicates": 0,
        "written": 100000,
        "bufferSize": 0,
        "batches": 15698
    },
    "totalQueryTimeMs": 900
}
```

500/500ms

```
==========================================
         Statistical Analysis Report
==========================================
Total Duration:      122141 ms
Messages Sent:       100000
Messages Received:   100000
Messages Failed:     0
Delivery Rate:       100.0%
Total Throughput:    818.73 msg/sec
------------------------------------------
Mean Latency:        4184.80 ms
Median Latency:      4059 ms
95th % Latency:      7760 ms
99th % Latency:      8169 ms
Min Latency:         55 ms
Max Latency:         8223 ms
------------------------------------------
Message Distribution:
  TEXT:  33361
  JOIN:  33482
  LEAVE: 33157
------------------------------------------
Throughput Per Room:
  Room  1: 40.31 msg/sec
  Room  2: 41.30 msg/sec
  Room  3: 41.07 msg/sec
  Room  4: 42.61 msg/sec
  Room  5: 39.18 msg/sec
  Room  6: 39.96 msg/sec
  Room  7: 40.54 msg/sec
```

```
[ec2-user@ip-172-31-29-109 ~]$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "totalMessages": 100000,
    "activeUsersLastHour": 62950,
    "messagesByType": [
        {
            "_id": "JOIN",
            "count": 33482
        },
        {
            "_id": "TEXT",
            "count": 33361
        },
        {
            "_id": "LEAVE",
            "count": 33157
        }
    ],
    "topUsers": [
        {
            "_id": "34764",
            "messageCount": 10
        },
        {
            "_id": "50207",
            "messageCount": 7
        },
        {
            "_id": "60107",
            "messageCount": 7
        },
        {
            "_id": "58209",
            "messageCount": 7
        },
        {
            "_id": "77553",
            "messageCount": 7
        },
        {
            "_id": "32186",
            "messageCount": 6
        },
        {
            "_id": "37526",
            "messageCount": 6
        },
        {
            "_id": "37201",
```

```
    ],
    "topRooms": [
        {
            "_id": "9",
            "messageCount": 5274
        },
        {
            "_id": "4",
            "messageCount": 5204
        },
        {
            "_id": "15",
            "messageCount": 5180
        },
        {
            "_id": "19",
            "messageCount": 5129
        },
        {
            "_id": "20",
            "messageCount": 5104
        },
        {
            "_id": "10",
            "messageCount": 5093
        },
        {
            "_id": "14",
            "messageCount": 5054
        },
        {
            "_id": "12",
            "messageCount": 5054
        },
        {
            "_id": "2",
            "messageCount": 5044
        },
        {
            "_id": "3",
            "messageCount": 5016
        }
    ],
    "room1MessageCount": 4923,
    "dbWriter": {
        "duplicates": 0,
        "written": 100000,
        "bufferSize": 0,
        "batches": 16501,
        "failed": 0
    },
    "totalQueryTimeMs": 853
}
```

1000/500ms

```
==========================================
         Statistical Analysis Report
==========================================
Total Duration:      122204 ms
Messages Sent:       100000
Messages Received:   100000
Messages Failed:     0
Delivery Rate:       100.0%
Total Throughput:    818.30 msg/sec
------------------------------------------
Mean Latency:        4078.28 ms
Median Latency:      3915 ms
95th % Latency:      7770 ms
99th % Latency:      8011 ms
Min Latency:         59 ms
Max Latency:         8092 ms
------------------------------------------
Message Distribution:
  TEXT:  33093
  JOIN:  33160
  LEAVE: 33747
------------------------------------------
Throughput Per Room:
  Room  1: 42.31 msg/sec
  Room  2: 39.47 msg/sec
  Room  3: 42.09 msg/sec
  Room  4: 40.13 msg/sec
  Room  5: 41.30 msg/sec
  Room  6: 40.33 msg/sec
```

```
[ec2-user@ip-172-31-29-109 ~]$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "totalMessages": 100000,
    "activeUsersLastHour": 63217,
    "messagesByType": [
        {
            "_id": "JOIN",
            "count": 33160
        },
        {
            "_id": "TEXT",
            "count": 33093
        },
        {
            "_id": "LEAVE",
            "count": 33747
        }
    ],
    "topUsers": [
        {
            "_id": "2892",
            "messageCount": 8
        },
        {
            "_id": "96068",
            "messageCount": 8
        },
        {
            "_id": "72733",
            "messageCount": 7
        },
        {
            "_id": "76762",
            "messageCount": 7
        },
        {
            "_id": "46134",
            "messageCount": 7
        },
        {
            "_id": "39777",
            "messageCount": 7
        },
        {
            "_id": "17557",
            "messageCount": 7
        },
        {
            "_id": "42924",
            "messageCount": 6
        },
        {
            "_id": "55157",
```

```
    "topRooms": [
        {
            "_id": "19",
            "messageCount": 5189
        },
        {
            "_id": "1",
            "messageCount": 5170
        },
        {
            "_id": "3",
            "messageCount": 5144
        },
        {
            "_id": "12",
            "messageCount": 5099
        },
        {
            "_id": "10",
            "messageCount": 5063
        },
        {
            "_id": "5",
            "messageCount": 5047
        },
        {
            "_id": "17",
            "messageCount": 5044
        },
        {
            "_id": "16",
            "messageCount": 5040
        },
        {
            "_id": "15",
            "messageCount": 5024
        },
        {
            "_id": "20",
            "messageCount": 5009
        }
    ],
    "room1MessageCount": 5170,
    "dbWriter": {
        "duplicates": 0,
        "written": 100000,
        "bufferSize": 0,
        "batches": 15373,
        "failed": 0
    },
    "totalQueryTimeMs": 994
}
```

5000/500ms

```
==========================================
         Statistical Analysis Report
==========================================
Total Duration:      122227 ms
Messages Sent:       100000
Messages Received:   100000
Messages Failed:     0
Delivery Rate:       100.0%
Total Throughput:    818.15 msg/sec
------------------------------------------
Mean Latency:        4608.81 ms
Median Latency:      4354 ms
95th % Latency:      9432 ms
99th % Latency:      9807 ms
Min Latency:         52 ms
Max Latency:         9855 ms
------------------------------------------
Message Distribution:
  TEXT:  33623
  JOIN:  32981
  LEAVE: 33396
------------------------------------------
Throughput Per Room:
  Room  1: 40.95 msg/sec
  Room  2: 41.59 msg/sec
  Room  3: 37.61 msg/sec
  Room  4: 40.96 msg/sec
  Room  5: 40.43 msg/sec
  Room  6: 42.18 msg/sec
  Room  7: 41.70 msg/sec
  Room  8: 41.07 msg/sec
```

```
[ec2-user@ip-172-31-29-109 ~]$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "totalMessages": 100000,
    "activeUsersLastHour": 63373,
    "messagesByType": [
        {
            "_id": "TEXT",
            "count": 33623
        },
        {
            "_id": "LEAVE",
            "count": 33396
        },
        {
            "_id": "JOIN",
            "count": 32981
        }
    ],
    "topUsers": [
        {
            "_id": "94848",
            "messageCount": 8
        },
        {
            "_id": "51311",
            "messageCount": 8
        },
        {
            "_id": "40350",
            "messageCount": 7
        },
        {
            "_id": "95753",
            "messageCount": 7
        },
        {
            "_id": "93844",
            "messageCount": 7
        },
        {
            "_id": "30997",
            "messageCount": 7
        },
        {
            "_id": "80421",
            "messageCount": 7
        },
        {
            "_id": "92958",
            "messageCount": 7
        },
        {
```

```
},
"topRooms": [
    {
        "_id": "17",
        "messageCount": 5534
    },
    {
        "_id": "13",
        "messageCount": 5299
    },
    {
        "_id": "15",
        "messageCount": 5183
    },
    {
        "_id": "6",
        "messageCount": 5155
    },
    {
        "_id": "7",
        "messageCount": 5097
    },
    {
        "_id": "2",
        "messageCount": 5084
    },
    {
        "_id": "9",
        "messageCount": 5081
    },
    {
        "_id": "14",
        "messageCount": 5044
    },
    {
        "_id": "8",
        "messageCount": 5020
    },
    {
        "_id": "10",
        "messageCount": 5017
    }
],
"room1MessageCount": 5005,
"dbWriter": {
    "duplicates": 0,
    "written": 100000,
    "bufferSize": 0,
    "batches": 14041,
    "failed": 0
},
"totalQueryTimeMs": 992
```

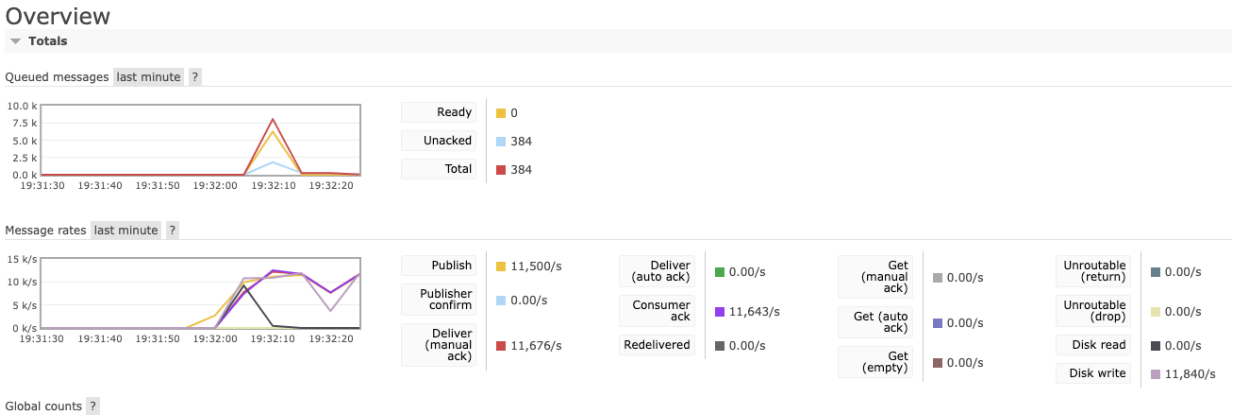
1000/100ms

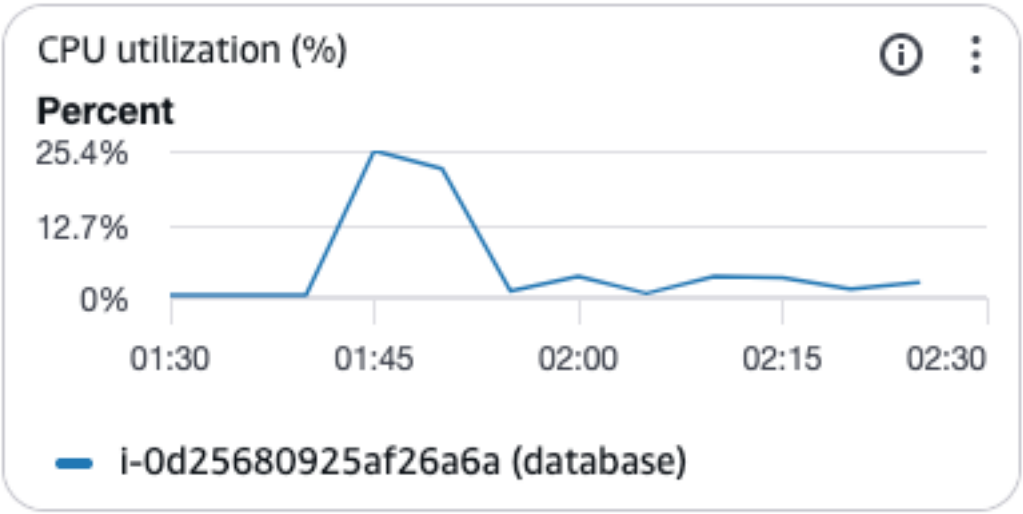
```
==========================================
         Statistical Analysis Report
==========================================
Total Duration:      122127 ms
Messages Sent:       100000
Messages Received:   100000
Messages Failed:     0
Delivery Rate:       100.0%
Total Throughput:    818.82 msg/sec
------------------------------------------
Mean Latency:        4512.26 ms
Median Latency:      4418 ms
95th % Latency:      8155 ms
99th % Latency:      8494 ms
Min Latency:         53 ms
Max Latency:         8559 ms
------------------------------------------
Message Distribution:
  TEXT:  33261
  JOIN:  33153
  LEAVE: 33586
------------------------------------------
Throughput Per Room:
  Room  1: 45.46 msg/sec
  Room  2: 39.75 msg/sec
  Room  3: 40.49 msg/sec
  Room  4: 40.13 msg/sec
  Room  5: 40.92 msg/sec
  Room  6: 41.86 msg/sec
  Room  7: 42.60 msg/sec
```

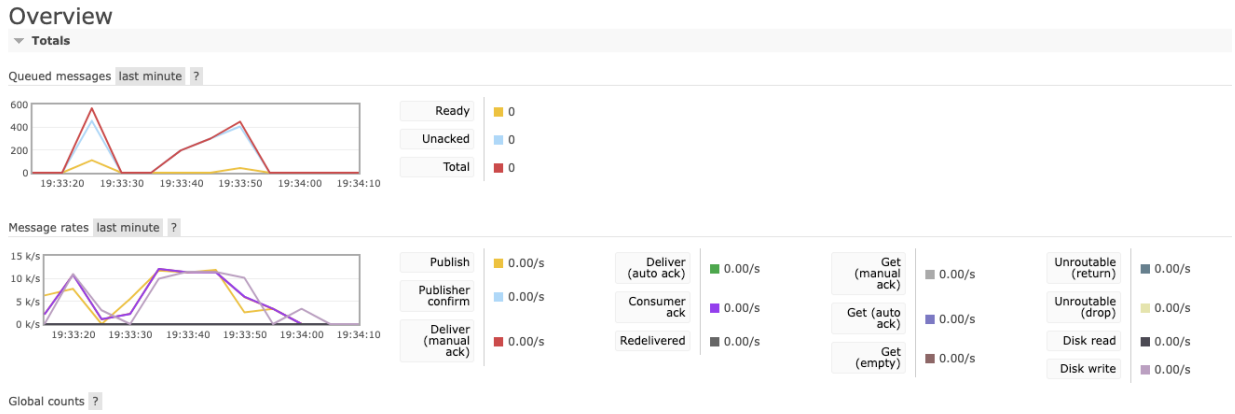
```
[ec2-user@ip-172-31-29-109 ~]$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "totalMessages": 100000,
    "activeUsersLastHour": 63111,
    "messagesByType": [
        {
            "_id": "TEXT",
            "count": 33261
        },
        {
            "_id": "LEAVE",
            "count": 33586
        },
        {
            "_id": "JOIN",
            "count": 33153
        }
    ],
    "topUsers": [
        {
            "_id": "35076",
            "messageCount": 8
        },
        {
            "_id": "53254",
            "messageCount": 8
        },
        {
            "_id": "92574",
            "messageCount": 8
        },
        {
            "_id": "42437",
            "messageCount": 7
        },
        {
            "_id": "98492",
            "messageCount": 7
        },
        {
            "_id": "62420",
            "messageCount": 7
        },
        {
            "_id": "34574",
            "messageCount": 7
        },
        {
            "_id": "30945",
            "messageCount": 7
        },
        {
            "_id": "41954",
```

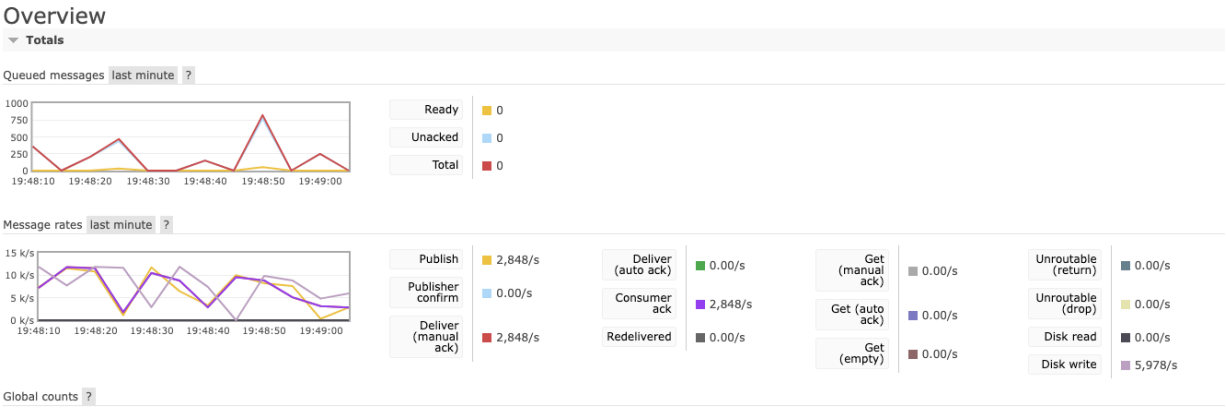
```
    "topRooms": [
        {
            "_id": "1",
            "messageCount": 5552
        },
        {
            "_id": "13",
            "messageCount": 5268
        },
        {
            "_id": "7",
            "messageCount": 5203
        },
        {
            "_id": "6",
            "messageCount": 5112
        },
        {
            "_id": "10",
            "messageCount": 5072
        },
        {
            "_id": "9",
            "messageCount": 5065
        },
        {
            "_id": "12",
            "messageCount": 5065
        },
        {
            "_id": "20",
            "messageCount": 5042
        },
        {
            "_id": "16",
            "messageCount": 5009
        },
        {
            "_id": "14",
            "messageCount": 5006
        }
    ],
    "room1MessageCount": 5552,
    "dbWriter": {
        "written": 100000,
        "bufferSize": 0,
        "batches": 14505,
        "failed": 0,
        "duplicates": 0
    },
    "totalQueryTimeMs": 977
}
```

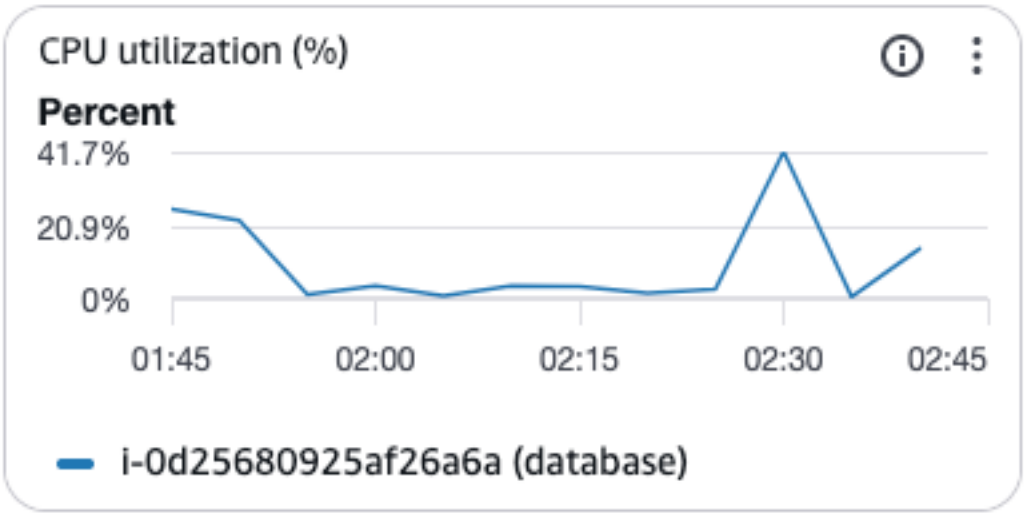
Test 3: Endurance Test:








Overview
Totals
Queued messages last minute ?
1000
750
500
250
0
19:48:10 19:48:20 19:48:30 19:48:40 19:48:50 19:49:00
Ready 0
Unacked 0
Total 0
Message rates last minute ?
15 k/s
10 k/s
5 k/s
0 k/s
19:48:10 19:48:20 19:48:30 19:48:40 19:48:50 19:49:00
Publish 2,848/s
Publisher confirm 0.00/s
Deliver (manual ack) 2,848/s
Deliver (auto ack) 0.00/s
Consumer ack 2,848/s
Redelivered 0.00/s
Get (manual ack) 0.00/s
Get (auto ack) 0.00/s
Get (empty) 0.00/s
Unroutable (return) 0.00/s
Unroutable (drop) 0.00/s
Disk read 0.00/s
Disk write 5,978/s
Global counts ?


CPU utilization (%)
Percent
41.7%
20.9%
0%
01:45
02:00
02:15
02:30
02:45
i-0d25680925af26a6a (database)

```
==========================================
         Statistical Analysis Report
==========================================
Total Duration:      249557 ms
Messages Sent:       3000000
Messages Received:   917514
Messages Failed:     0
Delivery Rate:       30.6%
Total Throughput:    3676.57 msg/sec
------------------------------------------
Mean Latency:        54862.31 ms
Median Latency:      50997 ms
95th % Latency:      116861 ms
99th % Latency:      120194 ms
Min Latency:         88 ms
Max Latency:         121317 ms
------------------------------------------
Message Distribution:
  TEXT:  306190
  JOIN:  306748
  LEAVE: 304576
------------------------------------------
Throughput Per Room:
  Room  1: 170.17 msg/sec
  Room  2: 204.39 msg/sec
  Room  3: 212.02 msg/sec
  Room  4: 159.89 msg/sec
  Room  5: 209.48 msg/sec
  Room  6: 181.84 msg/sec
  Room  7: 169.18 msg/sec
```

```
[ec2-user@ip-172-31-29-109 ~]$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "totalMessages": 917514,
    "activeUsersLastHour": 99983,
    "messagesByType": [
        {
            "_id": "JOIN",
            "count": 306748
        },
        {
            "_id": "TEXT",
            "count": 306190
        },
        {
            "_id": "LEAVE",
            "count": 304576
        }
    ],
    "topUsers": [
        {
            "_id": "58460",
            "messageCount": 24
        },
        {
            "_id": "69458",
            "messageCount": 24
        },
        {
            "_id": "12508",
            "messageCount": 24
        },
        {
            "_id": "44854",
            "messageCount": 23
        },
        {
            "_id": "78558",
            "messageCount": 23
        },
        {
            "_id": "49150",
            "messageCount": 23
        },
        {
            "_id": "44671",
            "messageCount": 22
        },
        {
            "_id": "42356",
            "messageCount": 22
        },
        {
```

```
],
"topRooms": [
    {
        "_id": "10",
        "messageCount": 54262
    },
    {
        "_id": "3",
        "messageCount": 52912
    },
    {
        "_id": "5",
        "messageCount": 52276
    },
    {
        "_id": "19",
        "messageCount": 52235
    },
    {
        "_id": "2",
        "messageCount": 51007
    },
    {
        "_id": "20",
        "messageCount": 49758
    },
    {
        "_id": "12",
        "messageCount": 48494
    },
    {
        "_id": "16",
        "messageCount": 47206
    },
    {
        "_id": "17",
        "messageCount": 46404
    },
    {
        "_id": "6",
        "messageCount": 45380
    }
],
"room1MessageCount": 42468,
"dbWriter": {
    "batches": 35759,
    "bufferSize": 0,
    "written": 917514,
    "duplicates": 0,
    "failed": 0
},
"totalQueryTimeMs": 9554
```